

E-commerce Funnel Analysis



An End-to-End Customer Journey & Conversion Analysis

Project Overview

What is Funnel Analysis?

Rapid global growth due to technology and internet access helps businesses understand how users progress through different stages of their purchasing journey and identifies where potential customers leave before completing a purchase.

What is Funnel Analysis?

- Analyzed customer sessions from an e-commerce platform.
- Built a complete funnel from browsing to purchase.
- Measured conversion rates and customer drop-offs.
- Identified performance across:
 - Marketing Channels
 - Regions
 - Devices
 - Product Categories
 - Time (Daily & Hourly)

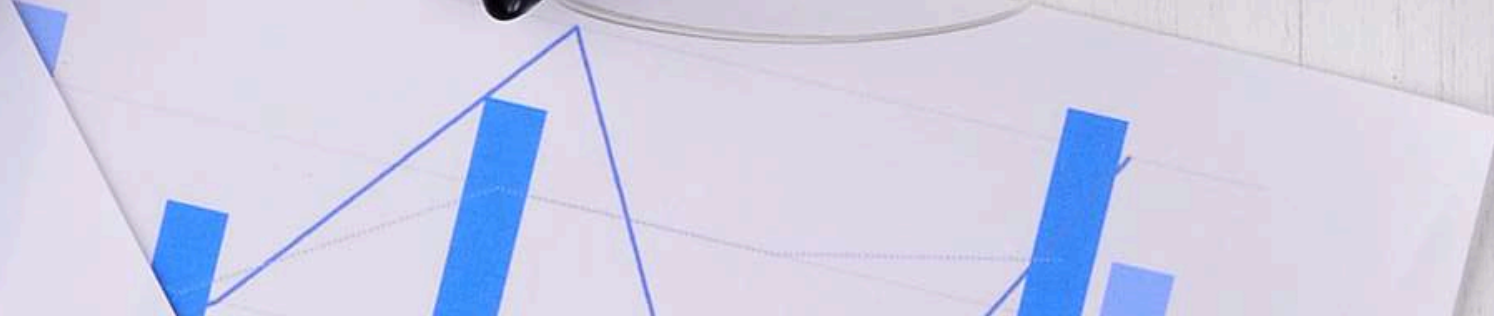
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Business Problem

The business wants to answer the following questions:

- Where are customers dropping out?
- Which marketing channels generate the highest conversions?
- Which regions perform best?
- Which devices lead to more purchases?
- Which product categories generate the highest revenue?
- At what time do customers convert the most?

Business decisions without these insights often result in:

- Poor marketing ROI
- Cart abandonment
- Low conversion rates
- Revenue loss



Tools & Technologiesals

The objective of this project is to:

Programming

- Python

Reporting

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly

Development

- Jupyter Notebook



Data Preprocessing & EDA

Data Cleaning & Feature Engineering

Performed:

- Missing value inspection
- Duplicate removal
- Datetime conversion
- Session aggregation
- Funnel stage creation
- Revenue aggregation
- Session duration calculation
- Daily and hourly feature extraction



Funnel Analysis Dashboard

Insights

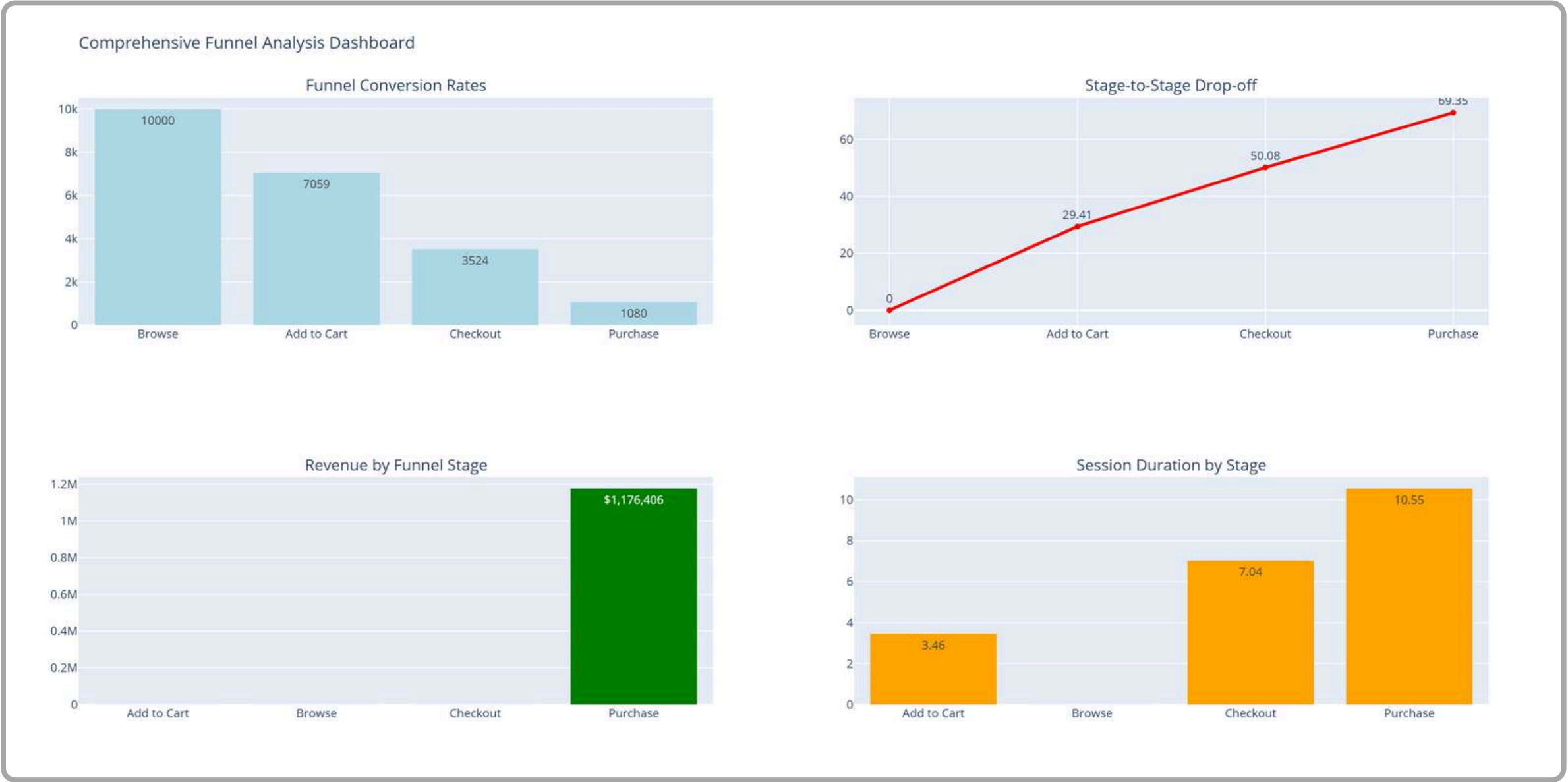
- This dashboard summarizes the complete customer journey.

Key Observations

- Overall funnel conversion decreases at every stage.
- Purchase stage shows the largest customer loss.
- Revenue accumulates mainly at the final stages.
- Longer sessions generally indicate stronger buying intent.

Business Impact

- Improving the highest drop-off stage can significantly increase conversions.



Channel Performance Analysis

Insights

Compared:

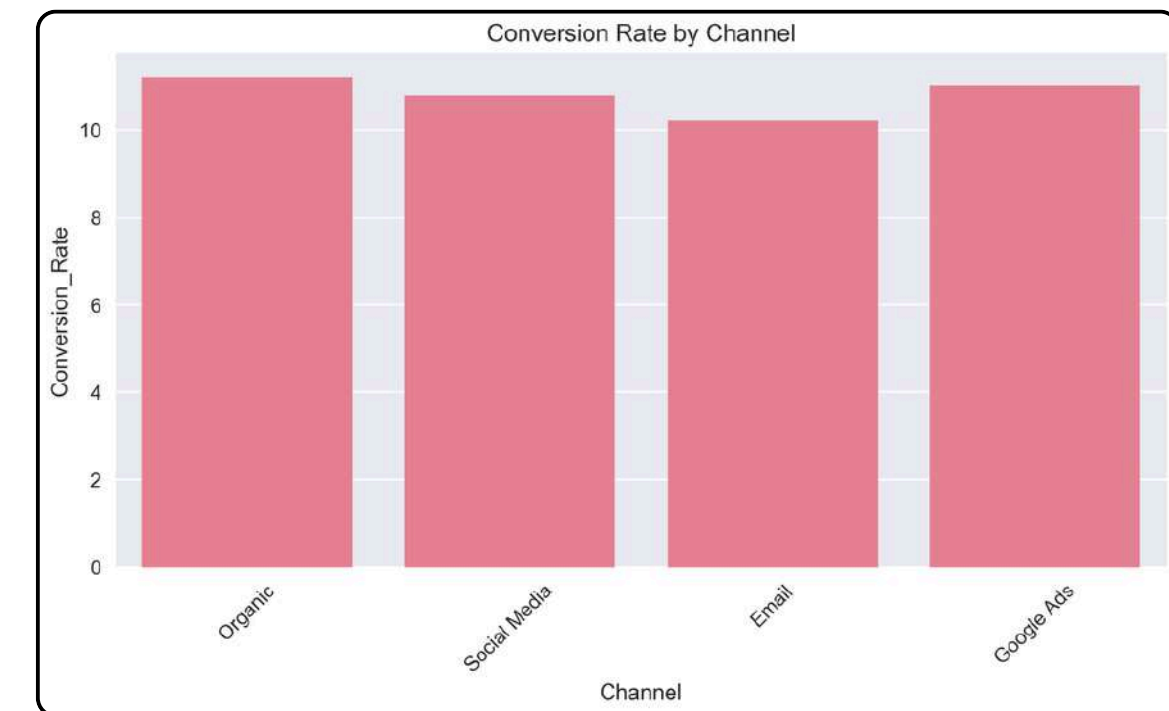
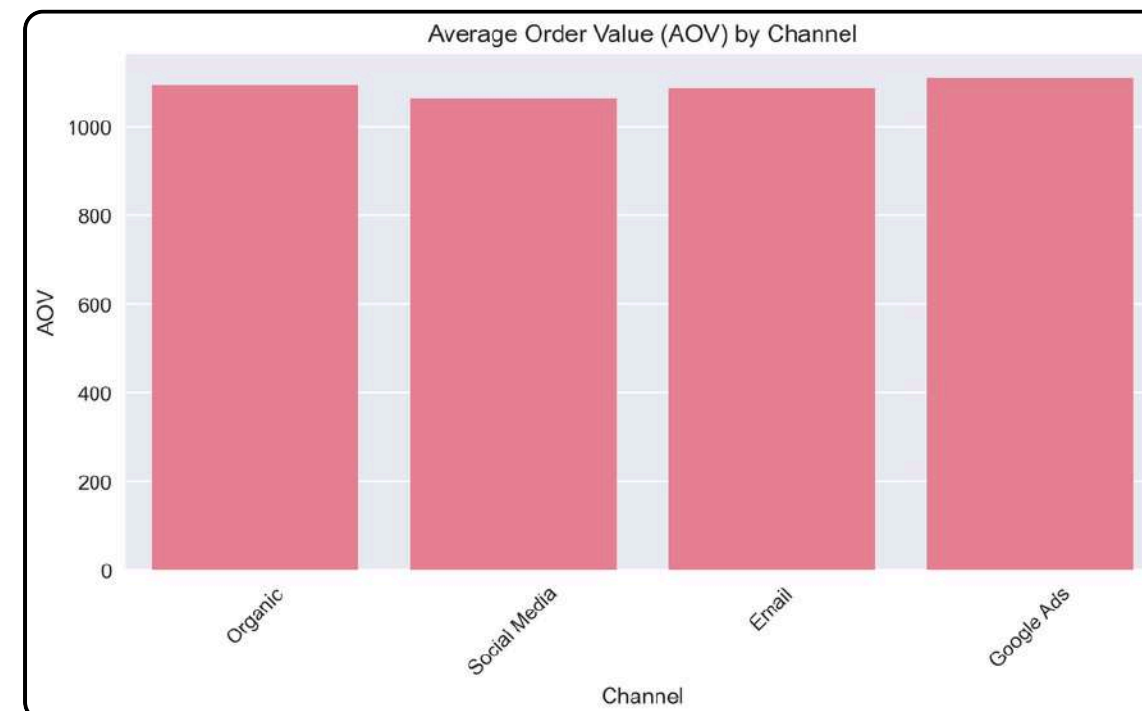
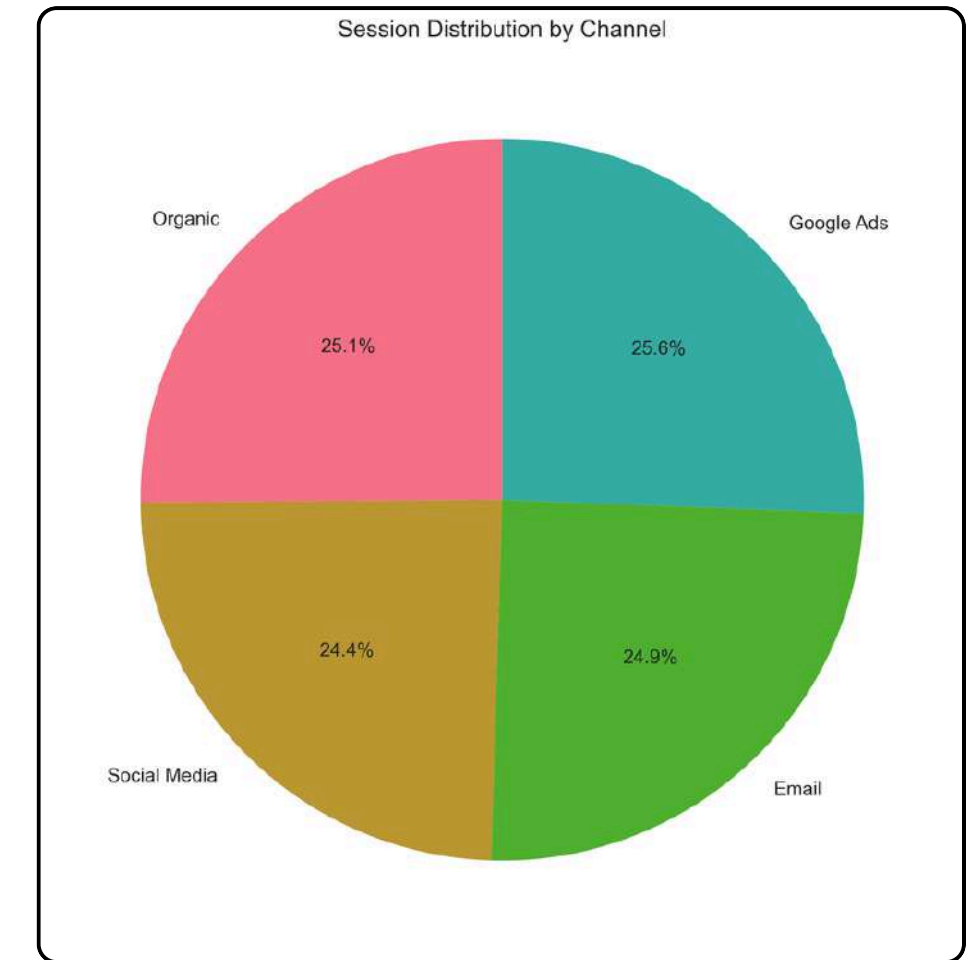
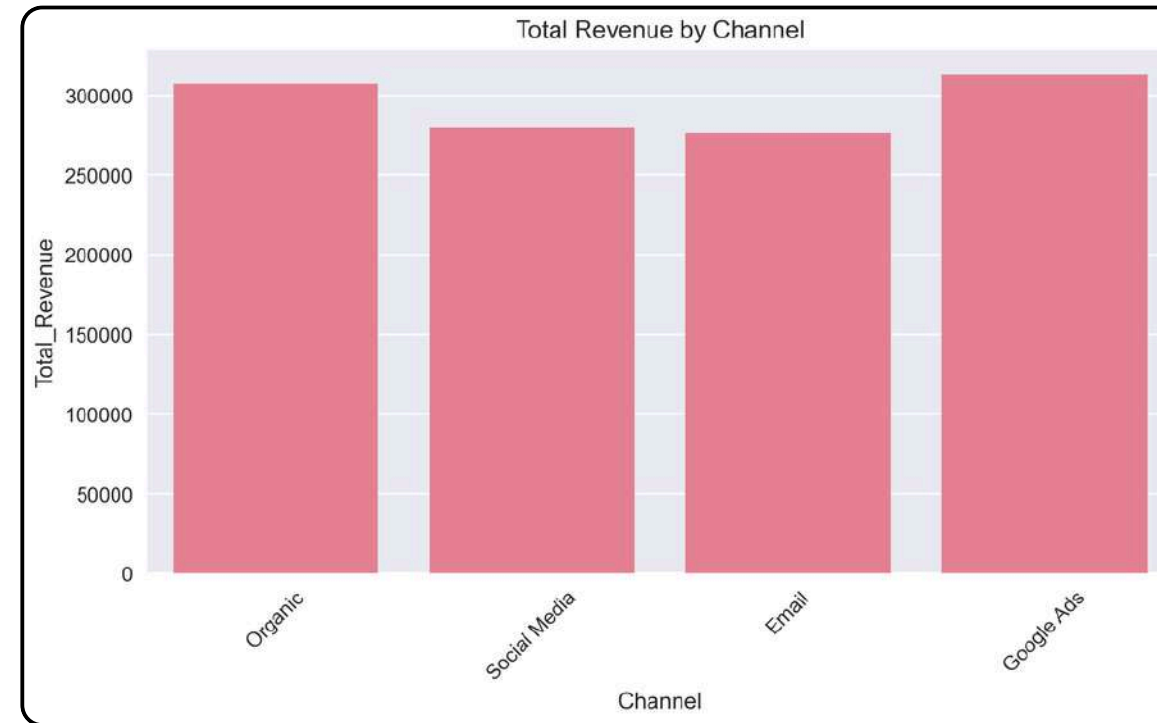
- Conversion Rate
- Revenue
- Average Order Value
- Session Distribution

Findings

- Some channels drive significantly higher conversions.
- High traffic does not always generate high revenue.
- Marketing budget should prioritize high-converting channels.

Business Recommendation

- Shift advertising investment toward the most profitable acquisition channels.



Regional Analysis

Insights

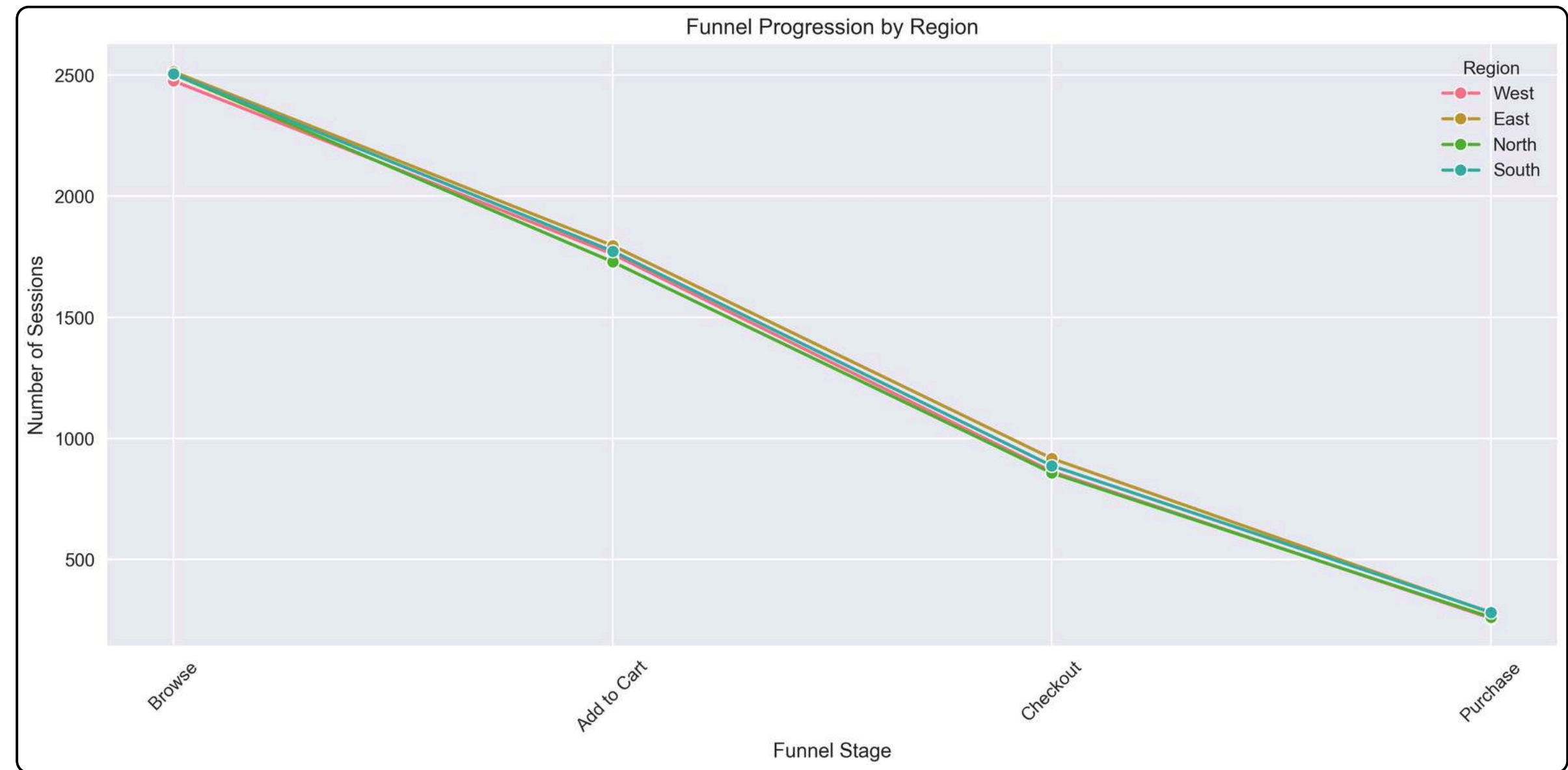
- Compared customer behavior across different regions.

Findings

- Some regions consistently outperform others.
- Regional conversion differences indicate varying customer preferences.
- Underperforming regions require localized marketing strategies.

Business Recommendation

- Optimize marketing campaigns based on regional customer behavior.



Stage-to-Stage Drop-off

Insights

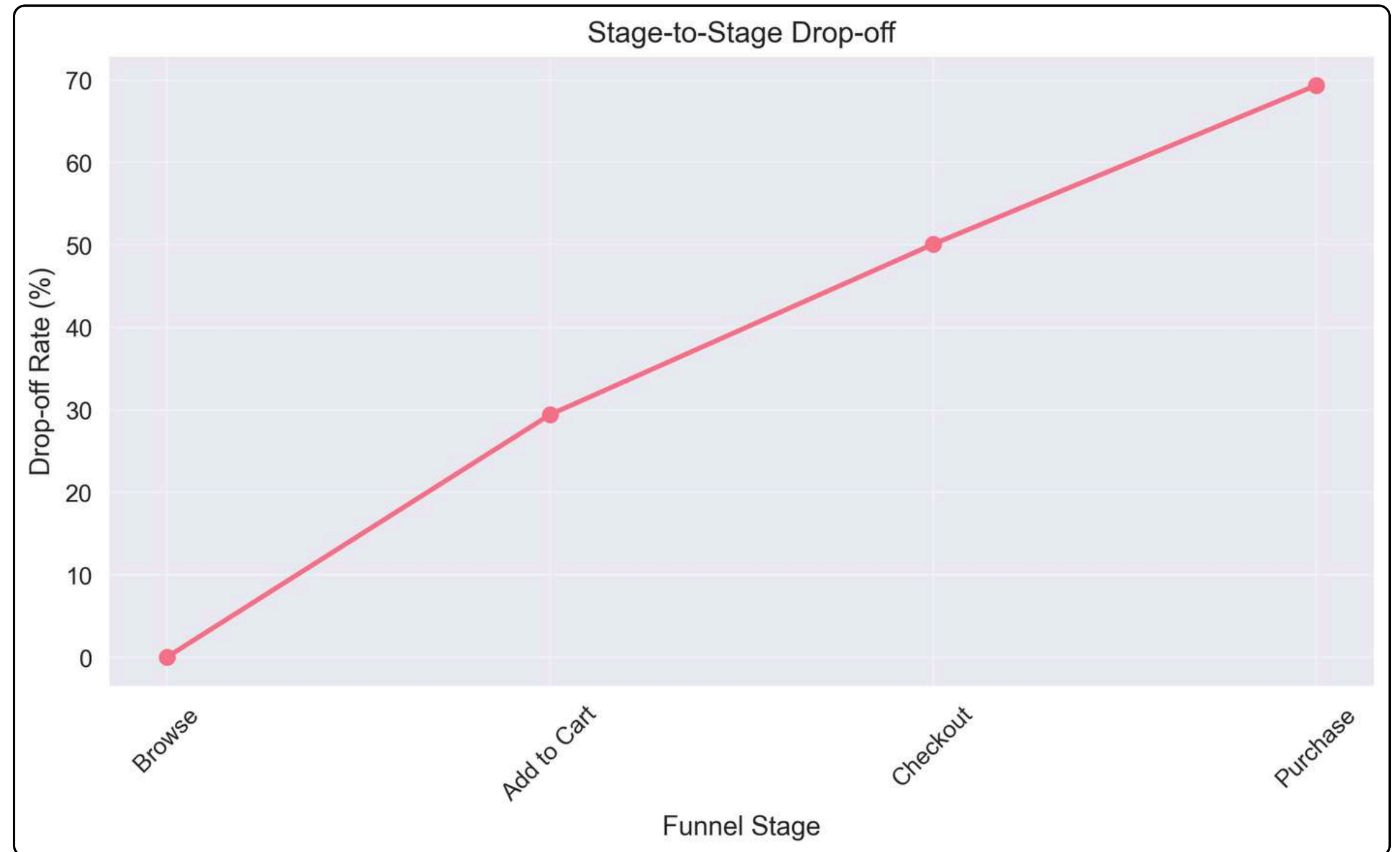
- Highest drop-off occurs before Purchase.
- Customer loss increases in later stages.

Findings

- Purchase stage has the largest drop-off (~69.35%).
- Checkout is the main conversion bottleneck.

Business Recommendation

- Optimize Simplify checkout.
- Use cart abandonment recovery campaigns.
- campaigns based on regional customer behavior.



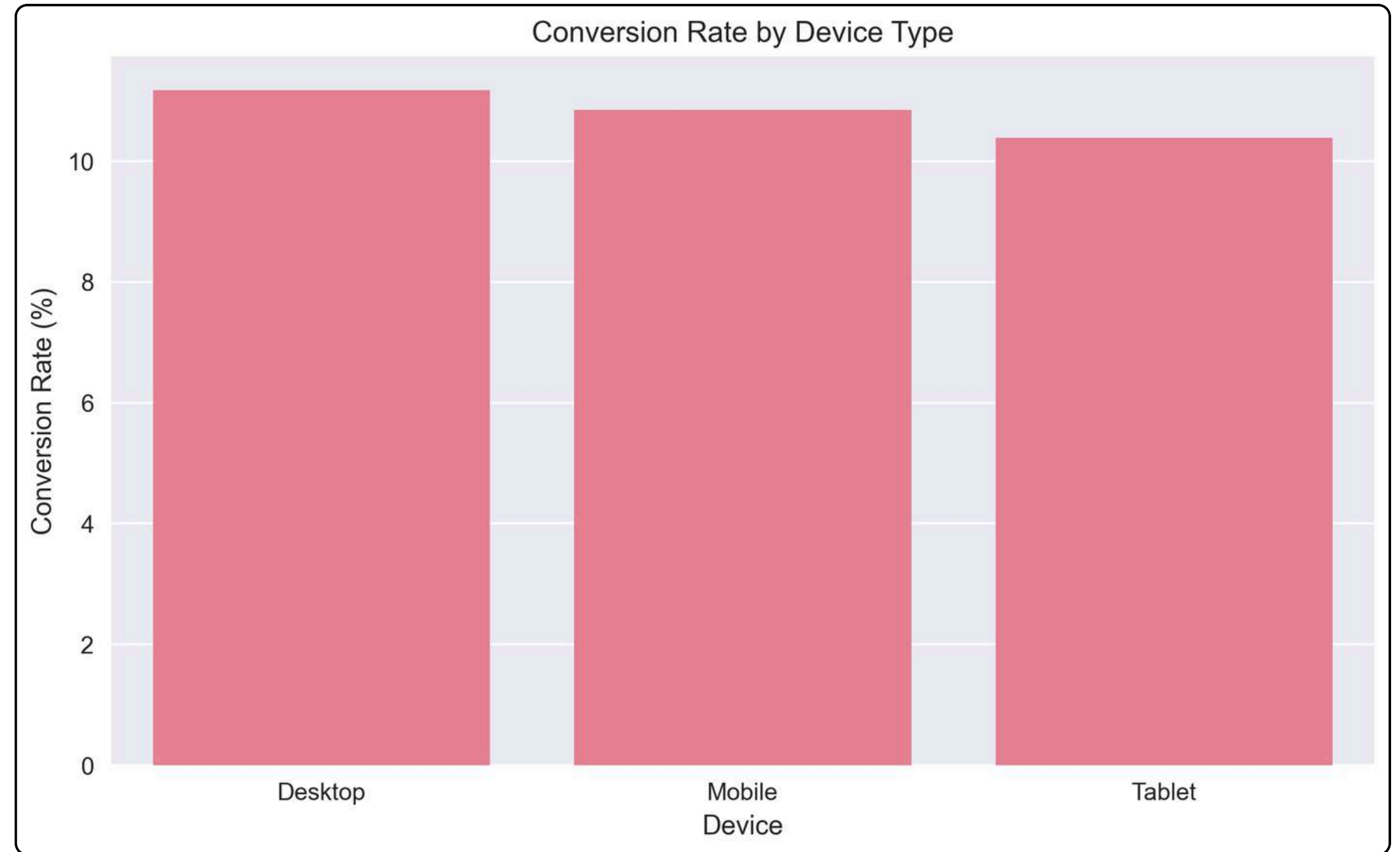
Device Performance

Insights

- Compared customer conversions by device.

Findings

- Device type significantly influences purchasing behavior.
- Mobile users may have lower conversion rates due to usability issues.
- Desktop users generally complete purchases more frequently.



Business Recommendation

- Improve the mobile checkout experience and website responsiveness.

Funnel Conversion Rate

Insights

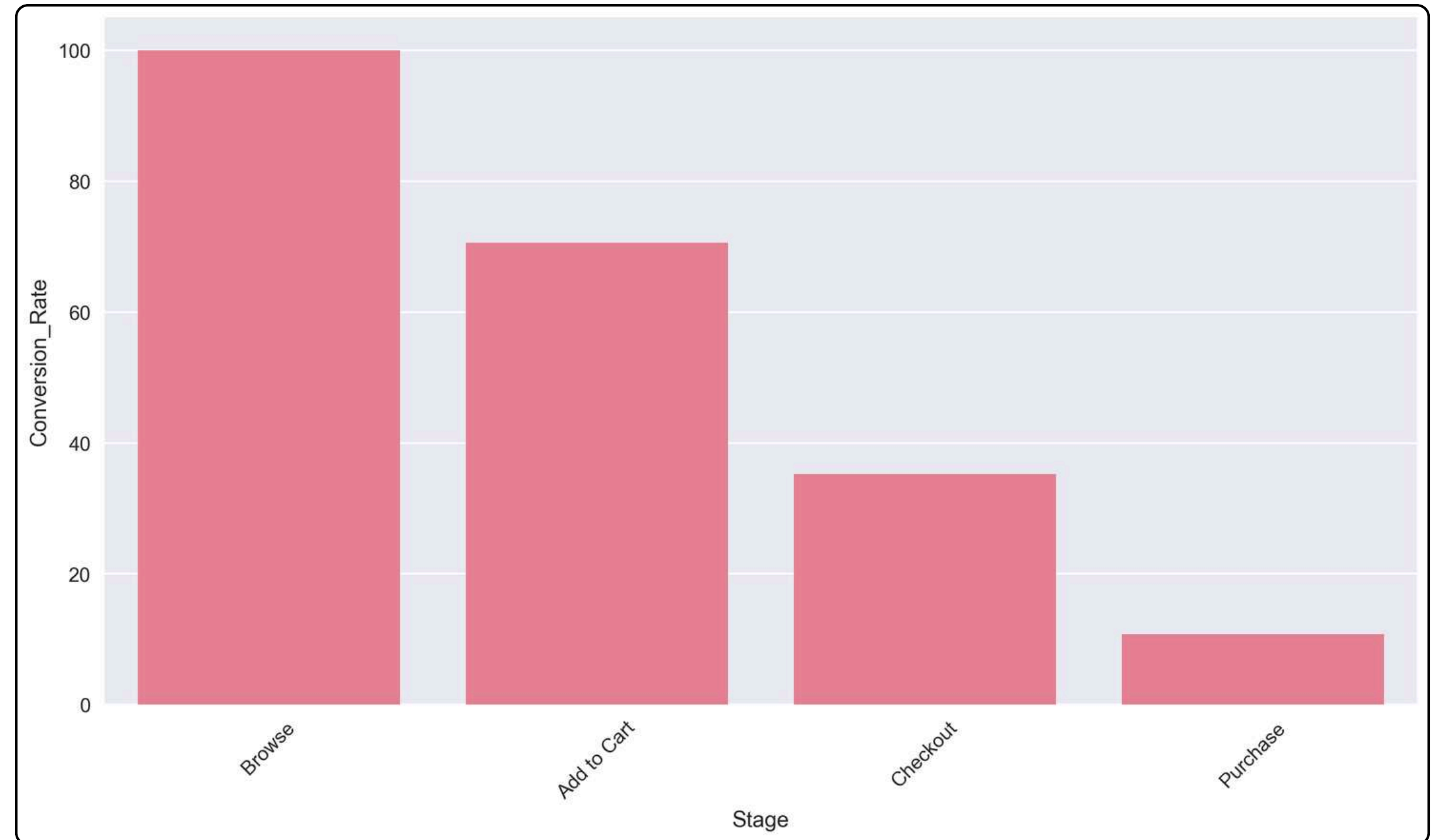
- Conversion decreases at each funnel stage.
- Only a small percentage of users complete a purchase.

Findings

- Customer drop-off occurs throughout the funnel.
- Improving stage-to-stage conversion can increase overall sales.

Business Recommendation

- Optimize key funnel stages.
- Simplify the customer journey to improve conversions.



Session Duration by Funnel Stage

Insights

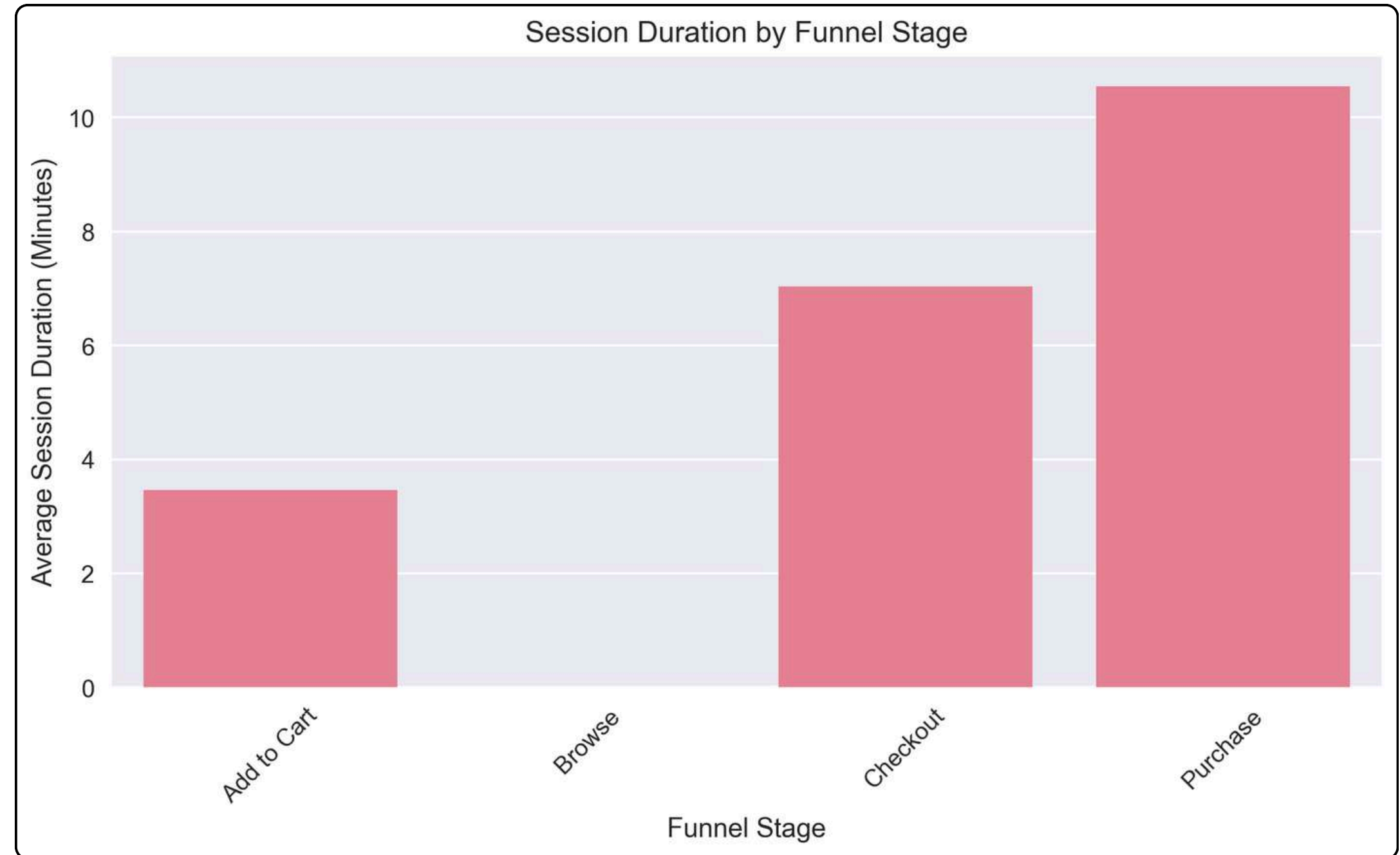
- Session duration varies across funnel stages.
- Users spend more time in the later stages of the funnel.

Findings

- Longer sessions indicate higher purchase intent.
- Short sessions may reflect low engagement or early exits.

Business Recommendation

- Improve user experience and streamline checkout.
- Optimize product pages to keep users engaged and increase conversions.



Product Category Analysis

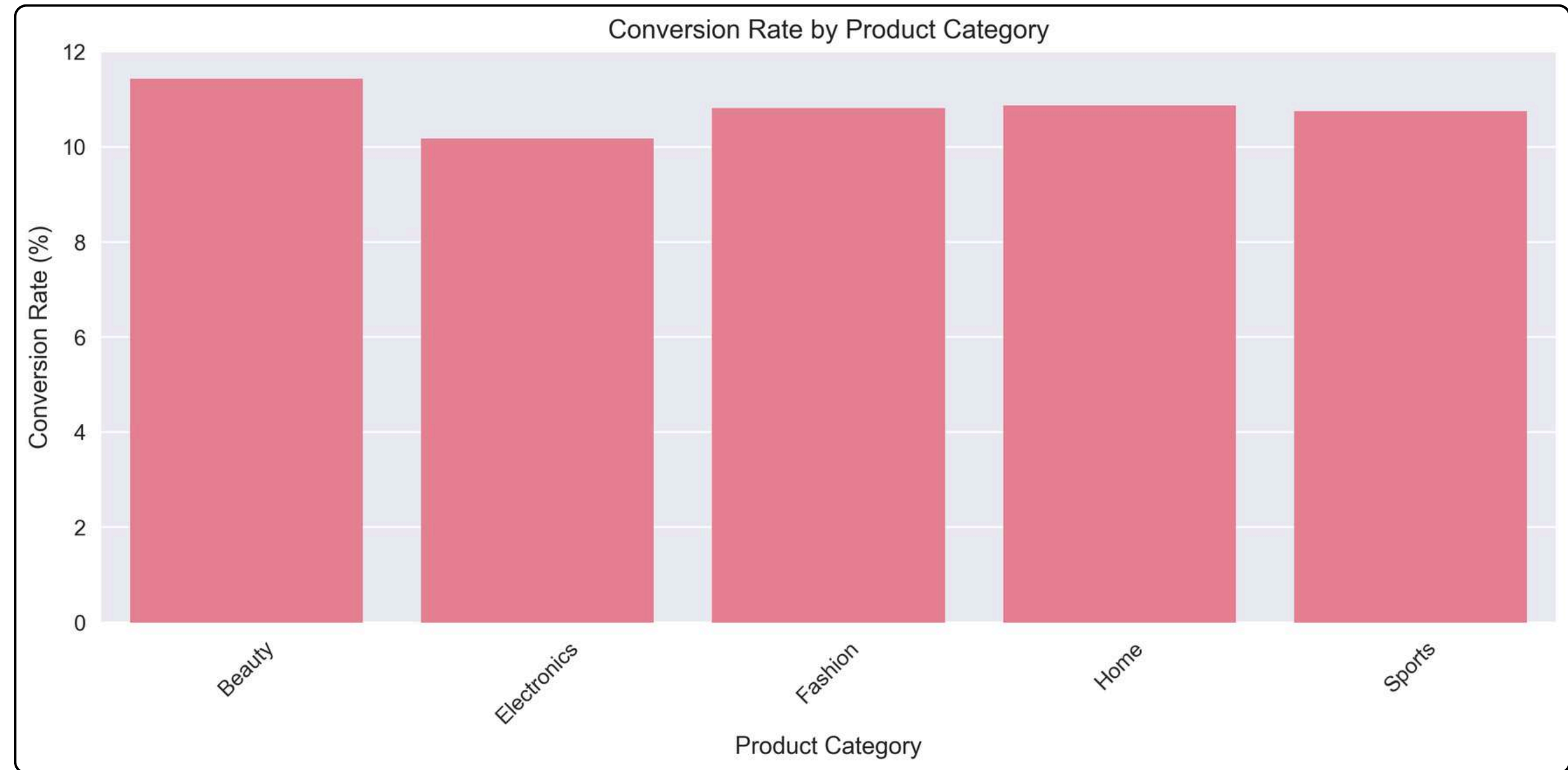
Insights

Measured:

- Conversion Rate
- Revenue
- Average Order Value

Findings

- Certain categories convert much better than others.
- High-revenue categories deserve increased promotional focus.
- Low-performing categories require pricing or merchandising improvements.



Daily Performance Trends

Insights

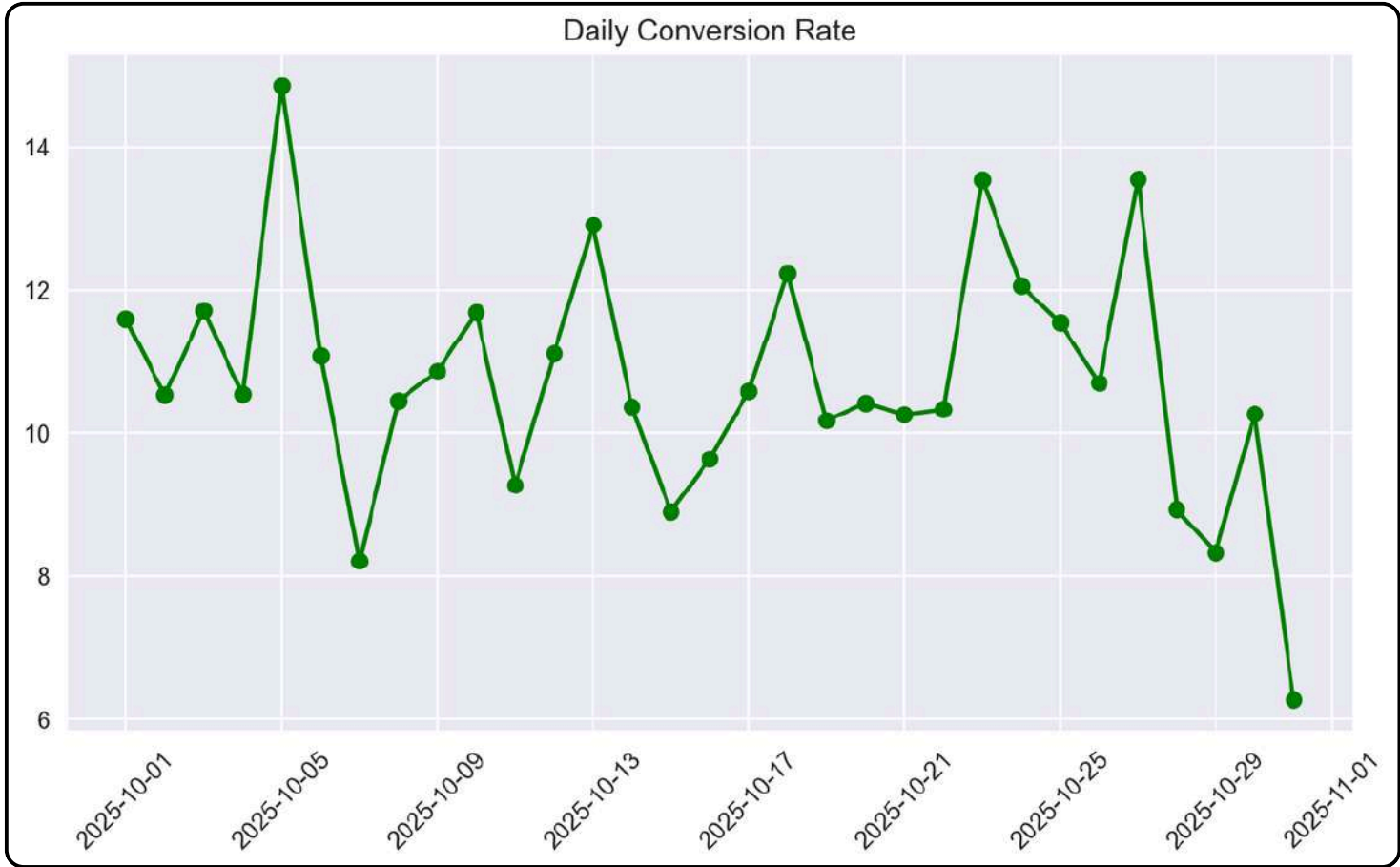
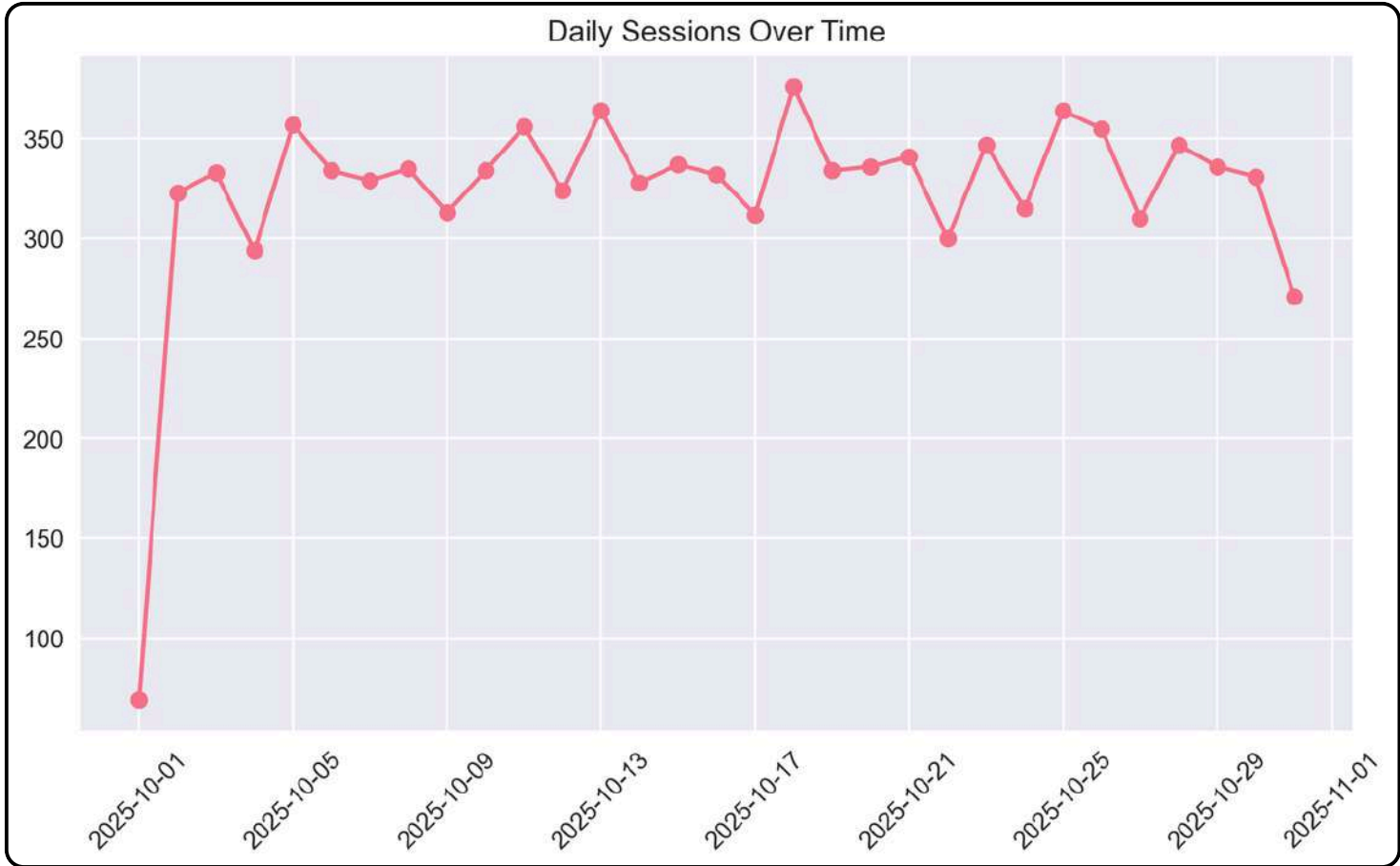
- Analyzed customer activity over time.

Findings

- Daily traffic fluctuates.
- Conversion rate varies by day.
- Certain days consistently outperform others.

Business Recommendation

- Schedule promotions during high-conversion periods.



Hourly Performance Trends

Insights

Measured:

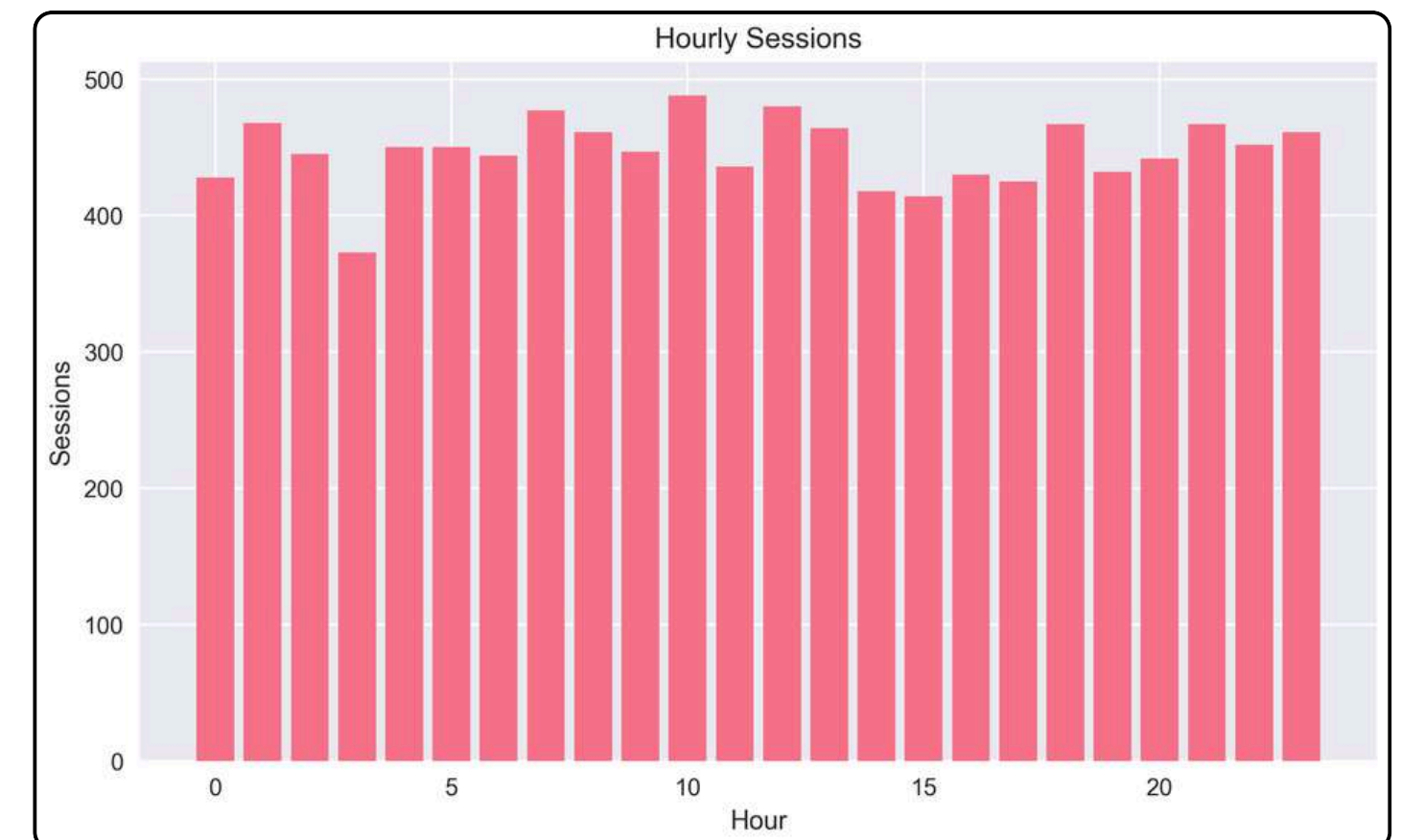
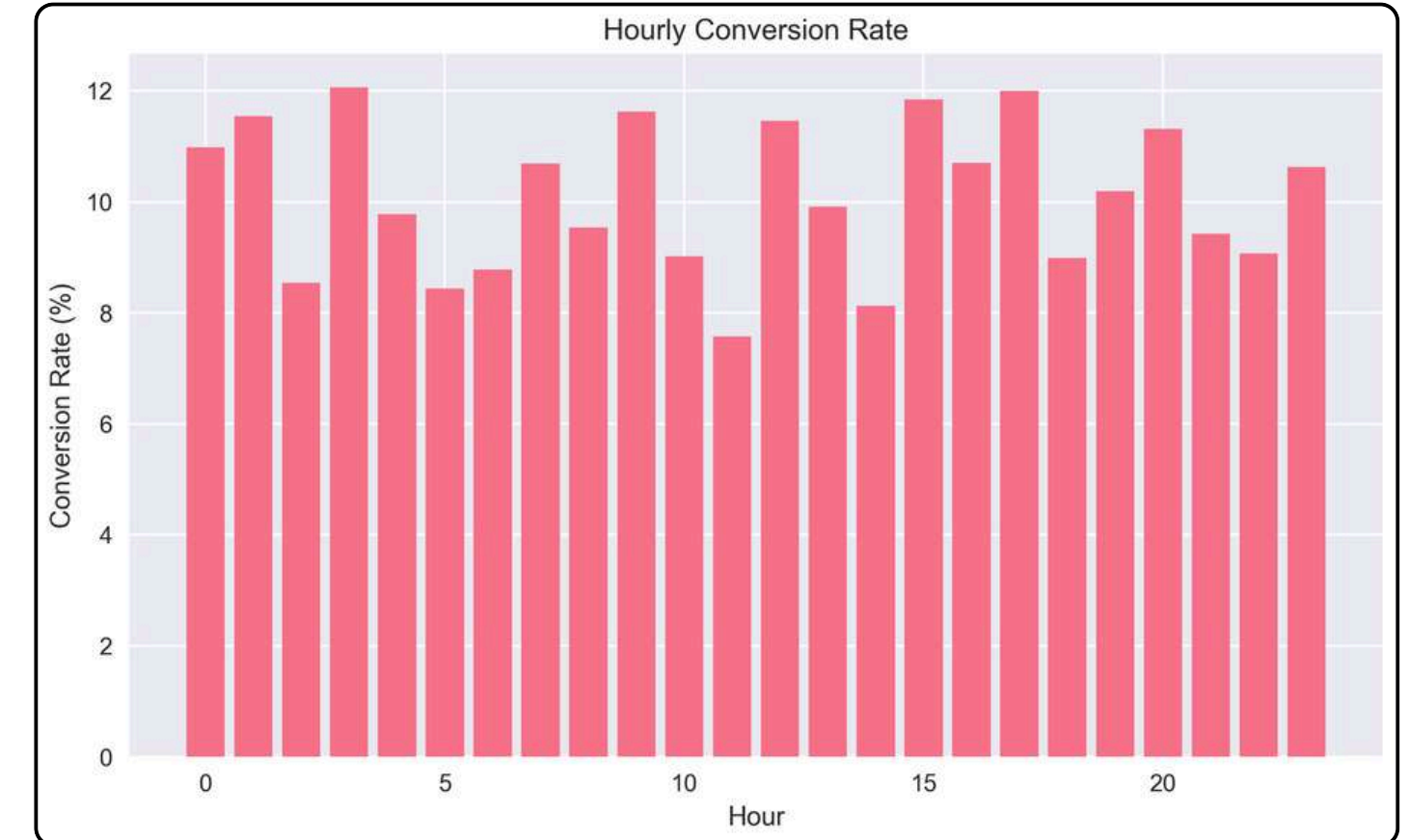
- Hourly sessions
- Hourly conversion rate

Findings

- Peak traffic hours differ from peak purchasing hours.
- High traffic doesn't always translate into purchases.

Business Recommendation

- Schedule advertisements and promotional campaigns during peak conversion hours rather than peak traffic hours.



Key Insights

Key Business Insights

- Largest customer drop-off occurs before purchase.
- Marketing channel performance varies significantly.
- Regional purchasing behavior differs.
- Device affects conversion.
- Product categories show different revenue potential.
- Customer activity changes throughout the day.



Business Recommendations

Strategic Recommendations

Funnel

- Improve checkout experience to reduce customer abandonment.

Regional

- Develop region-specific promotional campaigns.

Marketing

- Increase investment in the highest-converting channels.

Product

- Promote high-performing categories and optimize weak categories.

Device

- Enhance mobile user experience.

Time

- Launch campaigns during peak conversion hours.



Key Outcomes

Project Outcomes

Successfully developed an end-to-end analytical pipeline that:

- Cleaned and transformed raw customer data.
- Built customer funnel metrics.
- Generated 15 business visualizations.
- Produced Excel reports.
- Delivered actionable business recommendations.



Conclusion

The project demonstrates how exploratory data analysis and funnel analytics can uncover valuable customer behavior patterns.

- Key achievements include:
- Identification of customer drop-off stages.
- Performance comparison across channels and regions.
- Device and product category evaluation.
- Time-based behavioral analysis.
- Business-focused recommendations for improving conversions.





Thank You

bhaskarpal.official@gmail.com